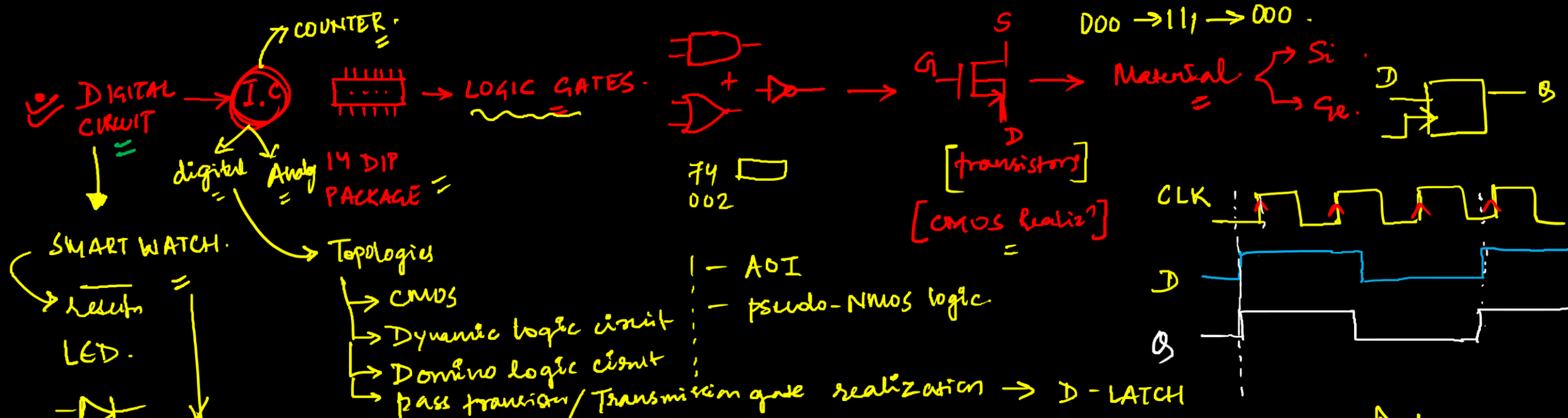




# SEMICONDUCTOR PHYSICS

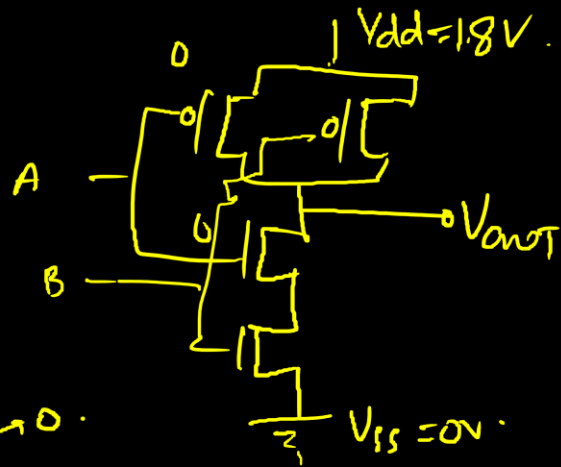
[19-03-2025]

Sailesh Mishra

RTL → synthesis (verification)  
register transfer level

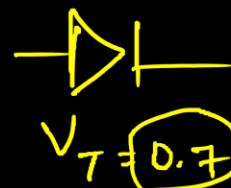
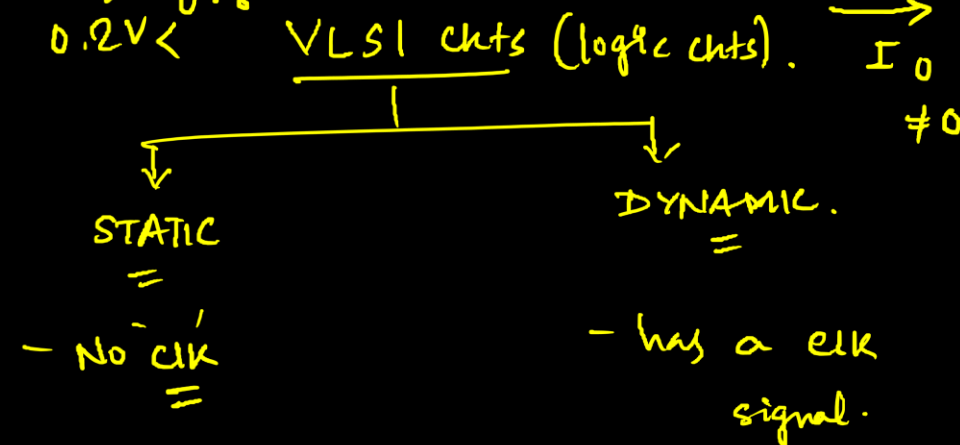
$$S = A \oplus B, C = A \cdot B$$

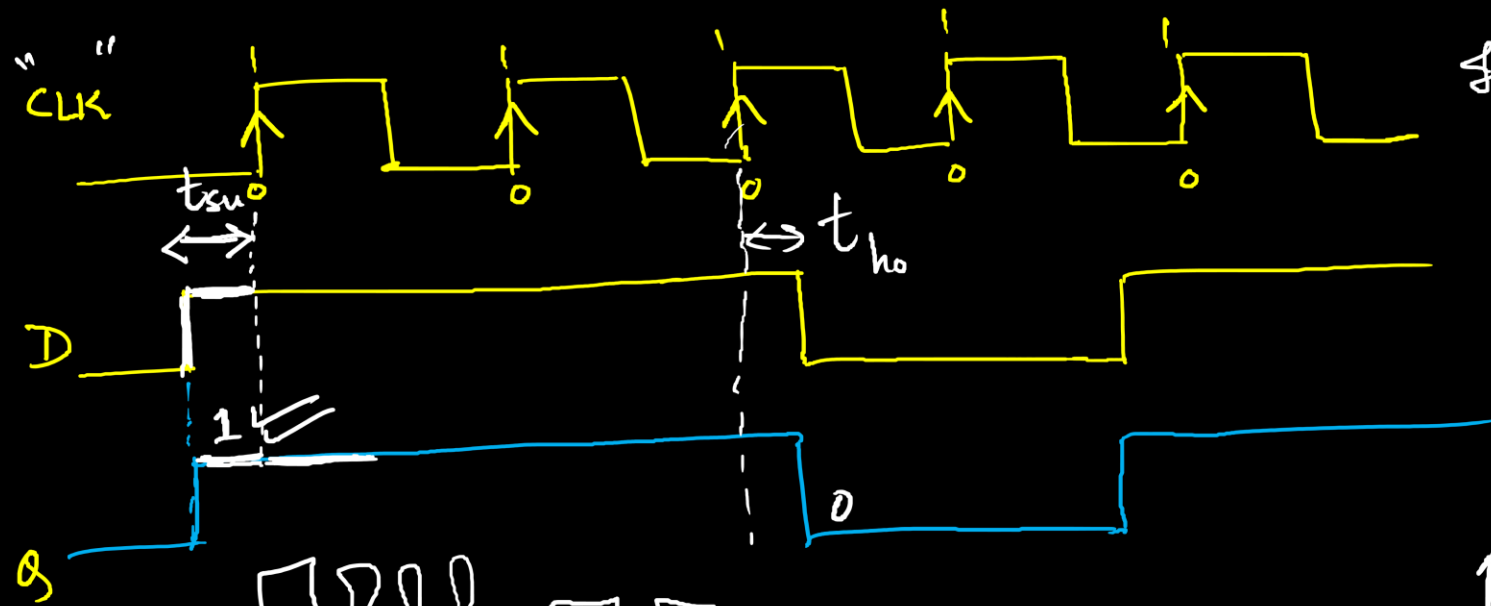
00 → 0  
01 → 1 10 → 1 11 → 0



$$V_{supply} > V_T \neq 0mA$$

0.2V < 0.6





$t_{su} = t_{clk}$   
 $t_{su} < t_{clk}$   
 { NO  $t_{su}$  violation }

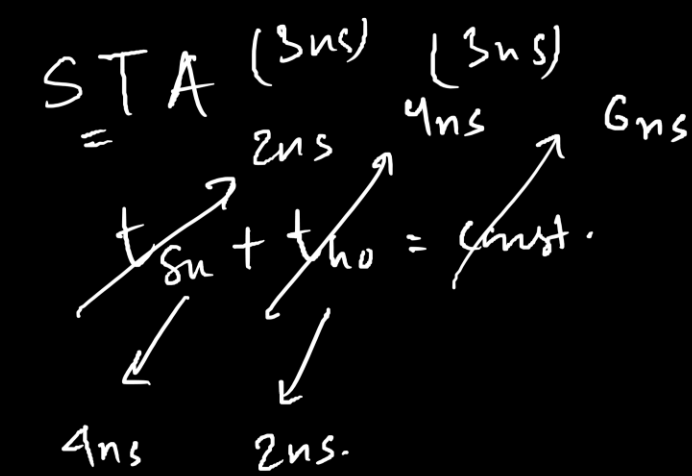


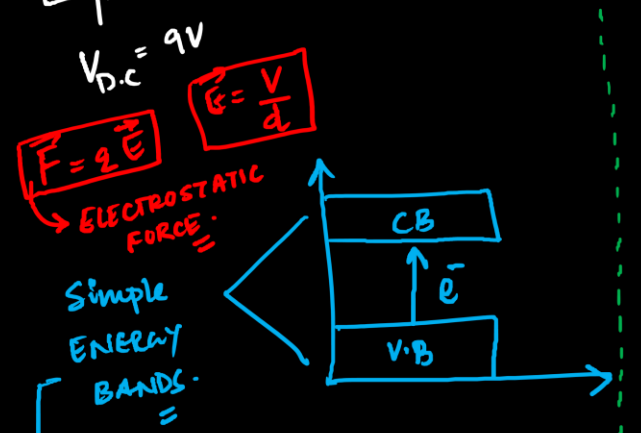
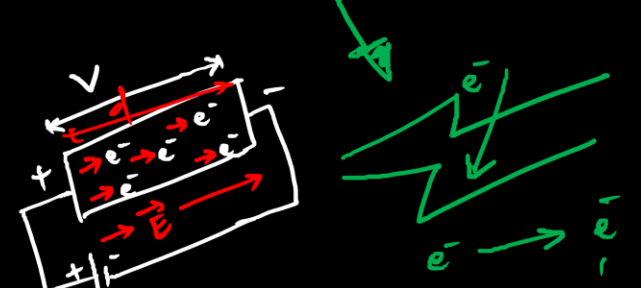
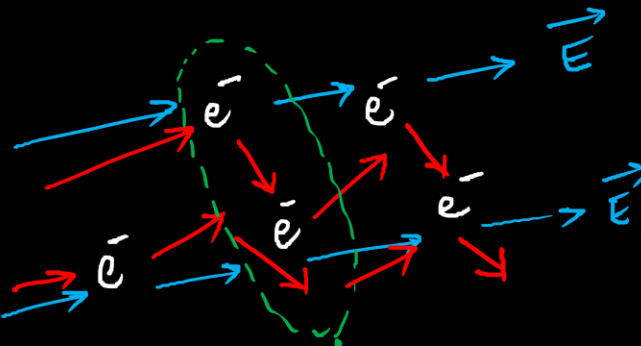
$t_{su} \propto \frac{1}{t_{ho}}$

SU.  $\rightarrow$  Railway Station  
 $T = \frac{1}{f}$

8:00  $\rightarrow$  8:00 am  
 7:10-7:20  $\rightarrow$  7:45

$t_{su}$ ; SETUP TIME.





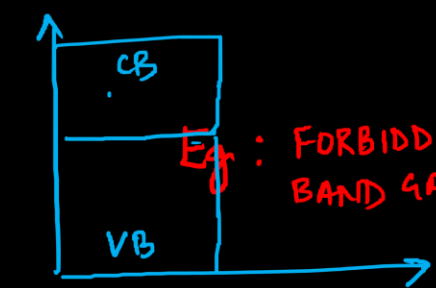
P.E  $\propto \frac{1}{\text{stability}}$

TE = PE + KE.  
 $= mgh + \frac{1}{2}mv^2$

# MATERIALS

## CONDUCTOR

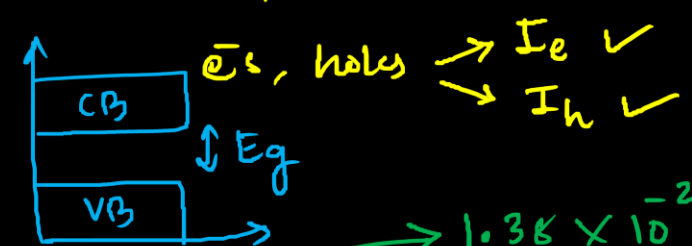
- uni polar Conduction  
 $(e^-) \rightarrow I \checkmark$



REMARK:-  
 The  $E_g$  in a Conductor is v. less !! So  
 Easy Conduction.

## SEMI-CONDUCTOR

- Bi-polar Conduction



$E_g$ : FORBIDDEN BAND GAP ENERGY  $\propto \frac{k_b \cdot T}{2e}$

$E_g = \frac{1.38 \times 10^{-23}}{1.6 \times 10^{-19}} \times \frac{I}{K} \times 300K$

$E_g = 25.87 \text{ V} \approx \boxed{26 \text{ mV}}$   
 $0.026 \text{ V}$

## INSULATOR

- NO CONDUCTION X

$1.38 \times 10^{-23} \text{ JK}^{-1}$

Absolute Temp (K)  $\rightarrow$  THERMAL ENERGY.  
 [room temp]

$27^\circ\text{C} \rightarrow 273\text{K} + 27^\circ\text{C} = 300\text{K}$

Thermally Generated Current  $\rightarrow I_e$   
 $\rightarrow I_h$

$26 \times 10^{-3} = 0.026$

$\Rightarrow$  Hence; It's NOT EASILY CONTROLLABLE or CALIBRABLE !!

# SEMI-CONDUCTOR

INSULATORS < Conductivity < CONDUCTORS

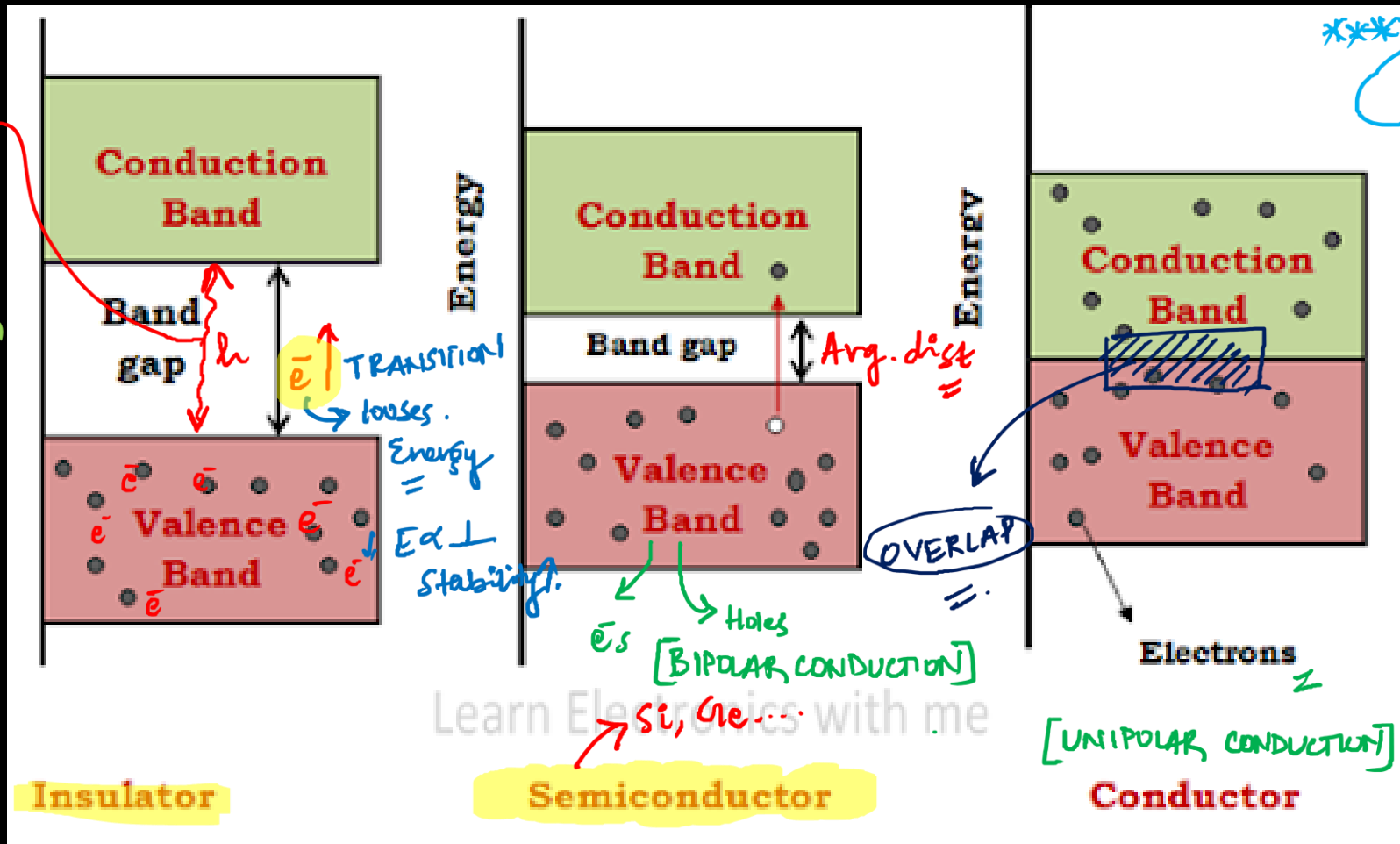
∴ This  $h$  is a distance, is a Material dependent property !!

if  $h \uparrow$  PE  $\uparrow$  TE  $\uparrow$  (E  $\uparrow$ , stability  $\downarrow$ )

\*\*\* dist  $\uparrow$  E  $\uparrow$  Cond.  $\downarrow$

$$E_g = \frac{1.38 \times 10^{-24} \text{ J/K} \times 300 \text{ K}}{1.6 \times 10^{-19}}$$

$$E_g = 2.5875 \times 10^{-3} \approx 25.87 \text{ mV} \approx \boxed{26 \text{ mV}}$$



DRIFT VELOCITY =  $\frac{1}{2} m v_d^2$

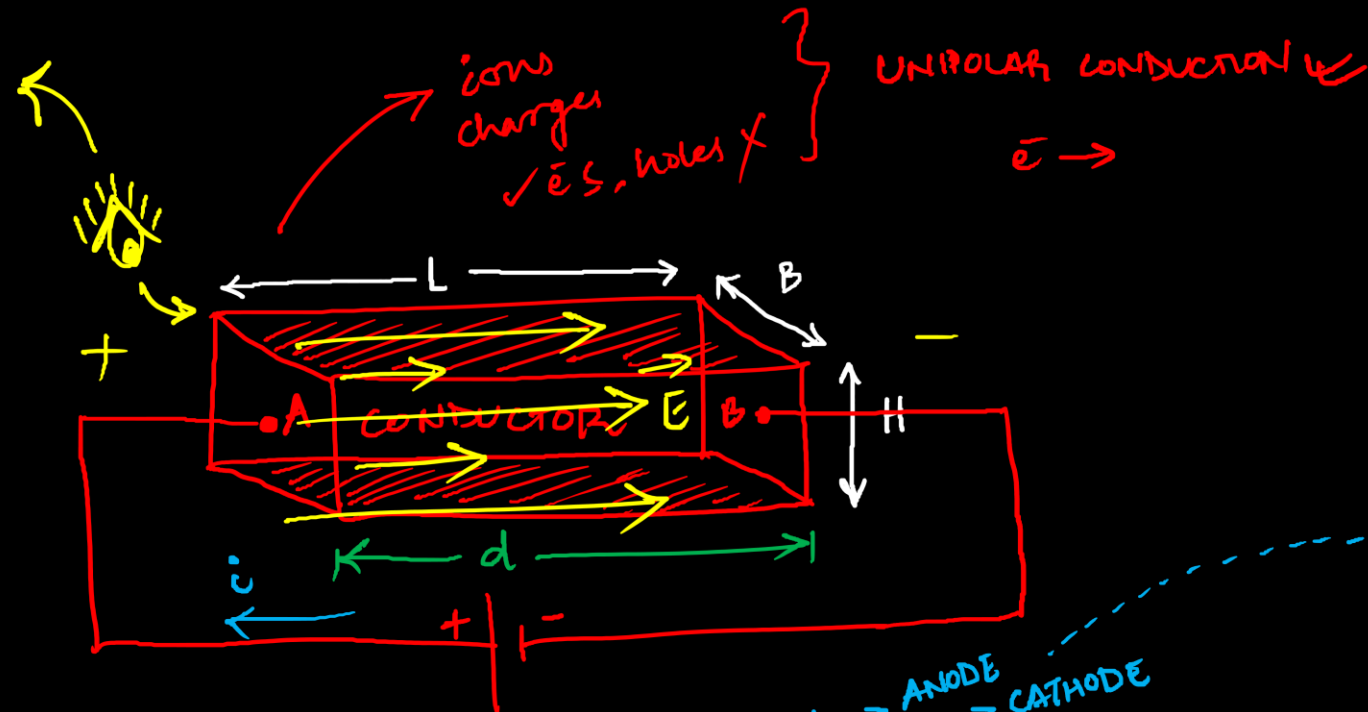
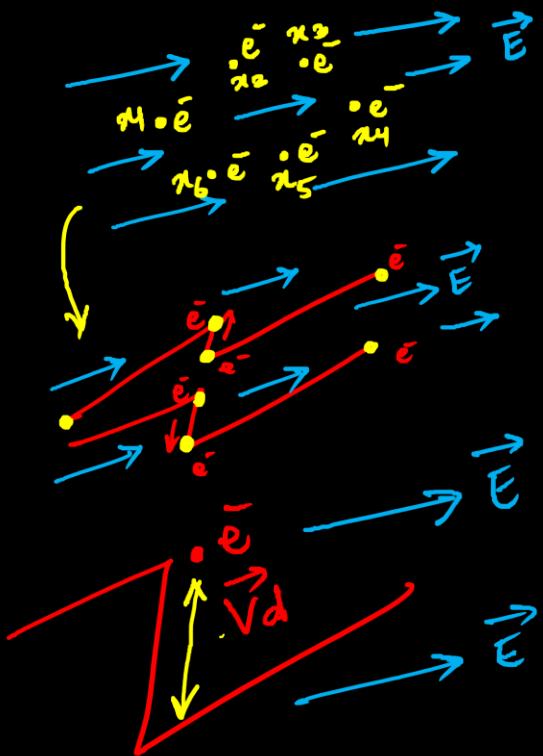
distance  $\uparrow$  TE = PE + KE  $\uparrow$  mgh  
The Energy required to Make a transition from VB  $\rightarrow$  CB is w/as BGEL.  
[Band Gap Energy level]

• It is directly proportional to  $\frac{k_b T}{2}$

where  $k_b \rightarrow$  Boltzman Constant,  $T =$  Absolute Temp. =  $27^\circ\text{C} = 273 + 27^\circ\text{C} = 300\text{K}$ .  
 $= 1.38 \times 10^{-24} \text{ J/K}$   $2 -$  charge of  $e^- = 1.6 \times 10^{-19} \text{ C}$

Learn Electronics with me

Si, Ge...



"9V Battery"



$$\vec{F} = q\vec{E} = 1.6 \times 10^{-19} \text{ C} \times 31.4 \text{ N/Cm}^2$$

$$V_{DC} = V_{AB} = V_A - V_B : \text{Relative Potential difference}$$

$$= 9 - 0 \text{ V}$$

$$= 9 \text{ V}$$

BATTERY  
 emf (potential): ENERGY  $\uparrow$   
 $TE = PE + KE$

$\rightarrow$  V-I Characteristics.  
 $\rightarrow$  due to emf; We're getting a  $\neq \vec{v}$ .

$$18 - 9 = 9 \text{ V} !!$$

ANODE  $\rightarrow$  CATHODE

Electrolytes  
 [ions, charges].  
 $=$   
 emf (V)  $\rightarrow \vec{v}$

$\odot \vec{E} = \text{ENERGY} = \frac{1}{2} \epsilon_0 \cdot E^2 = \text{Sum of } \underline{PE} + \underline{KE} = \underline{Mgh} + \frac{1}{2} m v_d^2 = m_e \left[ gh + \frac{1}{2} \frac{V_d^2}{f} \right]$

$(9.1 \times 10^{-31} \text{ kg})$

Environment dependent

permittivity of FREE SPACE

$\epsilon_0$ : free space or Vacuum

$= 8.85 \times 10^{-12} \frac{C^2}{Nm^2}$

$\vec{E} = \frac{V_{dc}}{d} \propto \frac{1}{2} \epsilon_0 \cdot E^2$

$\rightarrow [V_{D.C} \propto \text{Energy of } \vec{E}]$

$\vec{E}_{coulomb} = \frac{K \cdot Q_1 Q_2}{r^2} = \frac{1}{4\pi \epsilon_0} \cdot \frac{Q_1 Q_2}{r^2} = \frac{1}{4\pi \epsilon_0 \cdot \epsilon_r} \cdot \frac{Q_1 Q_2}{r^2}$

Relative permittivity =

$\rightarrow \text{Air} = 1 \checkmark$

$\rightarrow \text{Minerals} = 3 \checkmark$

$\rightarrow \text{O.C} = 3-5 \checkmark$

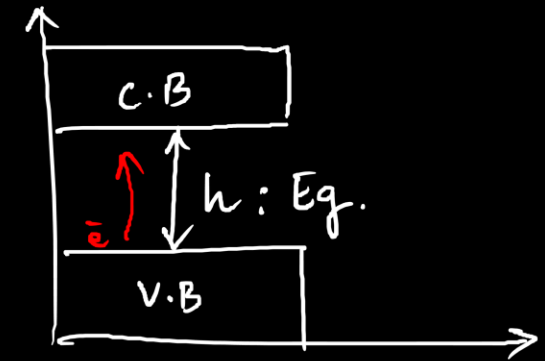
$\rightarrow \text{H}_2\text{O} = 79.6 \approx 80 \checkmark$

$\rightarrow \tau$  (Relaxation Time)  $\frac{0}{0}$  It's the time B/W any 2 successive collisions of  $e^-$ s inside the conductor...

$\vec{E} \uparrow \quad V_d \uparrow \quad \left( \text{speed} = \frac{\text{dist}}{\text{time}} \right) \quad t \downarrow \quad \tau \downarrow \quad \tau$  (Relaxation Time).

$I = n A e V_d$

CONDUCTIVITY :- [distance]<sub>(h)</sub> → Insulators > Semi-conductor > Conductors  
 Conductivity → Conductors > Semi-conductors > Insulators  
 'σ'



Why Semi-Conductors?

→ 0 EASIER CONTROLLABILITY. (Readily Available).

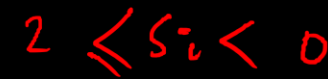
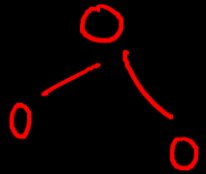
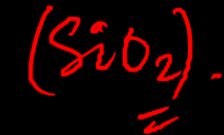
→ 0 LATTICE STRUCTURE.

→ 0 Thermally generated current.

→ 0 Cost Effective.

→ 0 Abundance In Availability.

→  $\hat{Si}$  (Sand): 2nd most Abundant Element/mineral found in Earth's crust.



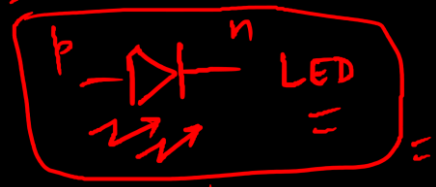
→ Tetrahedral.

"q = ne"

$i = \frac{dq}{dt} = \frac{d(ne)}{dt} = n \frac{dq}{dt} = \text{net}$

$n \uparrow \rightarrow i \uparrow$

# TYPES OF SEMI-CONDUCTORS



**INTRINSIC**

→ pure form / Natural  
 → These are NOT USED in VLSI Industry Mostly

Because they don't have good conductivity.

ELEMENTAL

Si, Ge

Group 14 elements

COMPOUND

GaAs, InP

**EXTRINSIC**

→ Intrinsic + DOPING !!

carriers /  $e^- \uparrow \uparrow \uparrow$   
 $I_d \uparrow \uparrow \uparrow$

**N-TYPE**

- Pentavalent impurities/dopants

- eg:- Sb, As, Bi, P  
 Group 15 elements

**P-TYPE**

Trivalent impurities/dopants.

eg- Al, B.  
 Group 13 elements

Need of doping ?? → To improve the conductivity of the device (MOSFET, BJT, JFET, pn junc)

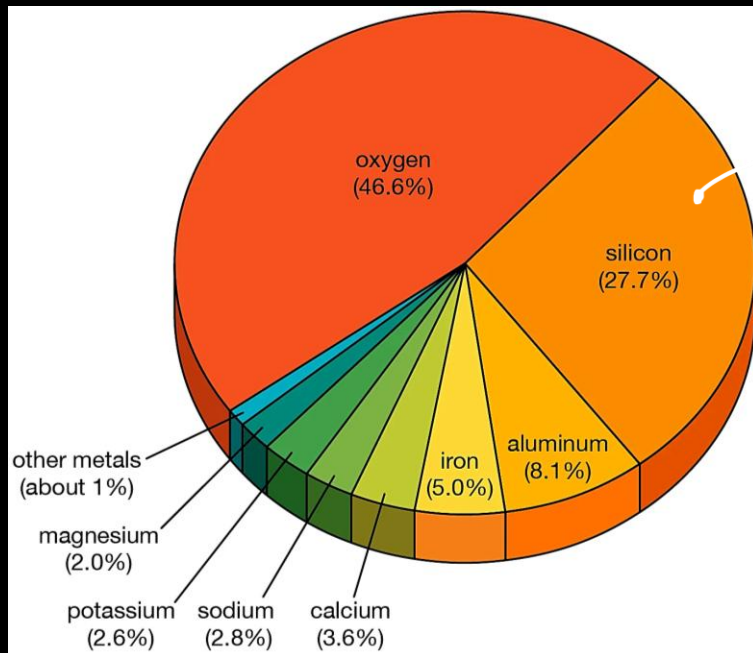
## Physical Properties of Silicon

- 1 Its atomic number is 14 with a relative atomic mass of 28.085 u.
- 2 It has a metal-like appearance and is brittle.
- 3 Silicon's density is approximately 2.33 grams per cubic centimeter.  
*→ Thermally Stable ✓*
- 4 Its melting point is 1410 degrees Celsius while its boiling point is 3265 degrees Celsius.
- 5 Silicon isotopes: -28, 29, and 30 are natural, but various artificial isotopes exist that are radioactive.
- 6 Its purest form is an intrinsic semiconductor.



## Chemical Properties of Silicon

- 1 Its chemical behavior is purely electropositive. ✓
- 2 Its chemical behavior is very similar to metals.
- 3 It does not easily integrate with oxygen or other related elements because it is solid.
- 4 It is relatively inactive at room temperature.
- 5 At higher temperatures, silicon is highly reactive.
- 6 In a molten state, it forms alloys.



→ 2nd MOST ABUNDANT Mineral found in Earth's crust ...

CHIPS (I.C) → ~~Si~~ → defence → Glacier -4°C  
-33°C  
-20°C.

Sahara 80-90°C, 62°C.



# MATERIALS BASED ON CONDUCTIVITY

